

Nonconvex Scenario Optimization for Data-Driven Reachability

Elizabeth Dietrich

Rosalyn Alex Devonport

Murat Arcak

Department of Electrical Engineering and Computer Sciences
University of California, Berkeley

EADIETRI@BERKELEY.EDU

ALEX_DEVONPORT@BERKELEY.EDU

ARCAK@BERKELEY.EDU

Abstract

Many of the popular reachability analysis methods rely on the existence of system models. When system dynamics are uncertain or unknown, data-driven techniques must be utilized instead. In this paper, we propose an approach to data-driven reachability that provides a *probabilistic* guarantee of correctness for these systems through nonconvex scenario optimization. We pose the problem of finding reachable sets directly from data as a chance-constrained optimization problem, and present two algorithms for estimating nonconvex reachable sets: (1) through the union of partition cells and (2) through the sum of radial basis functions. Additionally, we investigate numerical examples to demonstrate the capability and applicability of the introduced methods to provide nonconvex reachable set approximations.

Keywords: Reachability analysis, scenario optimization, data-driven methods

1. Introduction

To guarantee the safety of dynamical systems, *reachability analysis* is often used to determine the set of states that a system could possibly visit. However, in practice, computing exact reachable sets is an undecidable problem. For this reason, approximation methods are often used to reason about these systems and present guarantees. For example, over-approximated reachable sets guarantee safety when they do not overlap with unsafe regions of the state space.

There are many approaches that have been developed for this type of reachability analysis for systems with known dynamics. The most common of these include utilization of Hamilton-Jacobi differential equations (Mitchell et al., 2005; Bansal et al., 2017; Chen and Tomlin, 2018) or barrier certificates (Prajna, 2003; Prajna and Jadbabaie, 2004). While these techniques handle complex nonlinear dynamics well, their computational cost increases sharply with state dimensions. Set propagation techniques (Althoff, 2010; Althoff et al., 2021) iteratively compute a sequence of sets and achieve better scalability with state dimension. The most commonly used families of sets are ellipsoids (Kurzhanski and Varaiya, 2000; Botchkarev and Tripakis, 2000), hyperrectangles (Meyer et al., 2021), zonotopes (Girard, 2005), polytopes (Althoff et al., 2010), and support functions (Althoff and Frehse, 2016).

However, when the exact dynamics of a system are not known or only partially known, none of the techniques above can be used. Instead, we must estimate reachable sets in a data-driven manner. Several methods attempt to provide probabilistic guarantees of correctness for reachable sets directly from data. These methods include results from simulation and trajectory sensitivity analysis (Donzé and Maler, 2007; Girard and Pappas, 2006; Fan et al., 2017) or Gaussian processes (Devonport and Arcak, 2020a) and utilize simulation-based data to learn reachable sets. Other simulation-based and data-driven reachability methods include (Duggirala et al., 2013; Maidens

and Arcak, 2015; Arcak and Maidens, 2018; Lew and Pavone, 2020; Alanwar et al., 2021; Sun and Mitra, 2022; Qi et al., 2018).

Another data-driven approach to estimating reachable sets utilizes results from scenario optimization (Yang et al., 2017; Sartipizadeh et al., 2019; Devonport and Arcak, 2020b). This approach reduces the assumptions imposed on a system and can be applied to any system which admits simulation. Scenario optimization is an approach to solving chance-constrained optimization problems by solving a non-probabilistic relaxation of the original problem (Dembo, 1991). Scenario optimization has been used in solving robust control problems (Marseglia et al., 2014) and specifically problems related to reachability (Hewing and Zeilinger, 2020; Xue et al., 2020).

In this paper, we generalize the scenario-based reachability method of (Devonport and Arcak, 2020b). The scenario formalism therein is restricted to the convex case, which features critically in the construction of the probabilistic safety guarantees. However, this formalism places certain formal restrictions, such as convex parameterization, that preclude many popular classes of estimators. We generalize to a nonconvex formalism that allows for a broader class of sets. In particular, both the parametric representation of the minimal reachable set estimator and the reachable set itself can be nonconvex. Additionally, the existing work of (Devonport and Arcak, 2020b) yields a-priori complexity bounds for a desired probability of a problem. Our presented approach does not require a-priori bounds, as we calculate the probability of the original problem after solving the relaxed optimization problem. This allows us to solve a problem given any number of samples and significantly decreases the computational cost in finding reachable sets through scenario optimization.

We present two approaches in Section 3, both of which allow nonconvex reachable set approximations. In the first approach we examine the union of partition cells, and in the second approach we examine the sum of radial basis functions, and use sublevel sets as reachable set estimates.

2. Nonconvex Scenario Optimization

Take Δ to be a probability space with a σ -algebra and a probability measure $\mathbf{P}$, and let a *scenario*, δ , be a random outcome from this probability space. Since probability $\mathbf{P}$ is not known, it is not possible to directly compute the probability that an unseen scenario will violate a given set of constraints. Instead we use these scenarios, $\delta^{(i)}$, to construct a scenario optimization problem. Nonconvex scenario optimization (Campi et al., 2018; Garatti and Campi, 2024) is a technique to a-posteriori evaluate the robustness level of a scenario solution. Consider any constrained optimization problem of the form

$$\begin{aligned} & \underset{x \in \mathcal{X}}{\text{minimize}} && f(x) \\ & \text{subject to} && x \in \bigcap_{i=1, \dots, N} \mathcal{X}_{\delta^{(i)}} \end{aligned} \tag{1}$$

where $x \in \mathcal{X} \subseteq \mathbb{R}^d$ is the decision variable, $\mathcal{X}_{\delta^{(i)}}$ are constraints, and $\delta^{(1)}, \dots, \delta^{(N)}$ are N independently and identically-distributed (i.i.d) scenarios. There are no other restrictions on f and $\mathcal{X}_{\delta}$.

In solving (1), we aim to find a solution that is robust against constraint violation. The violation probability of a given $x \in \mathcal{X}$ is defined as $V(x) = \mathbf{P}\{\delta \in \Delta : x \notin \mathcal{X}_{\delta}\}$. Let x_N^* be the solution to (1) and define the violation of (1) to be $V(x_N^*)$. This is the probability that a new, randomly selected scenario, δ , will violate the constraints of (1). If $V(x_N^*) \leq \epsilon$, then (1) is robust against constraint violation at level ϵ . If the value of ϵ we achieve in our a-posteriori evaluation is not at the intended level, we iteratively increase N and recalculate ϵ . Therefore, this approach takes on a *wait-and-judge* perspective (Campi and Garatti, 2018).

We determine ϵ a-posteriori as a function of *support scenarios*. A scenario, δ , is a *support scenario* if its removal changes the solution of (1). We evaluate the number of support scenarios, s_N^* , by re-solving (1) upon individual removal of each scenario. If removing an individual scenario changes the solution to (1), then it is a support scenario. Through this process, we obtain an irreducible set of support scenarios with cardinality s_N^* . To calculate an estimate of ϵ based on s_N^* , $\epsilon(s_N^*)$, we first choose a confidence parameter, β , then calculate $\epsilon(s_N^*)$ through Theorem 1.

Theorem 1 ((Campi et al., 2018), Theorem 1) Given $\beta \in (0, 1)$, for any $s_N^* \in \{0, 1, \dots, N\}$, where N is the number of scenario samples, let

$$\epsilon(s_N^*) := \begin{cases} 1 & \text{if } s_N^* = N, \\ 1 - \sqrt[N-s_N^*]{\frac{\beta}{N(s_N^*)}}} & \text{otherwise.} \end{cases} \quad (2)$$

Then, the following probability bound holds:

$$\mathbf{P}\{V(x_N^*) > \epsilon(s_N^*)\} \leq \beta. \quad (3)$$

If we apply this general scenario theory to convex problems in which we restrict $\mathcal{X}_\delta$ from (1) to be a family of convex constraints, we can bound the number of support scenarios. We know $s_N^* < d$ where d is the number of optimization variables. It is known that a convex optimization problem with d optimization variables will never have more than d support scenarios. Therefore, we refine the definition of ϵ in Theorem 1 as follows:

$$\epsilon(s_N^*) := \begin{cases} 1 & \text{if } s_N^* \geq d, \\ 1 - \sqrt[N-s_N^*]{\frac{\beta}{d(s_N^*)}}} & \text{otherwise.} \end{cases} \quad (4)$$

3. Nonconvex Scenario-Based Reachability

We define a forward reachable set as $\mathcal{R} = \{\Phi(t_1; t_0, x_0, d) : x_0 \in \mathcal{X}_0, d \in \mathcal{D}\}$ where $\mathcal{X}_0 \subseteq \mathbb{R}^{n_x}$ is the set of initial states, $\mathcal{D}$ is the set of disturbance signals $d : [t_0, t_1] \rightarrow \mathbb{R}^{n_d}$, and $\Phi : \mathcal{X}_0 \times \mathcal{D} \rightarrow \mathbb{R}^{n_x}$ is the state transition function. This is the set of all states to which the system can transition to at time t_1 from state $\mathcal{X}_0$ at time t_0 subject to disturbances in $\mathcal{D}$. Since we cannot compute exact reachable sets, we aim to compute an approximation, $\mathcal{R}$, that is close to the true reachable set in a probabilistic sense.

Let $X_0 \in \mathcal{X}_0$ and $D \in \mathcal{D}$ be random variables, define $R = \Phi(t_1; t_0, X_0, D)$, and take accuracy parameter $\epsilon \in (0, 1)$ and confidence parameter $\beta \in (0, 1)$. Given a set of samples $\delta^{(i)} = \Phi(t_1; t_0, x_{0i}, d_i), i = 1, \dots, N$ where $x_{01}, \dots, x_{0N} \stackrel{i.i.d}{\sim} X_0, d_1, \dots, d_N \stackrel{i.i.d}{\sim} D$. We will explore reachable set estimates of the form

$$\mathcal{R}(\theta) = \{x \in \mathbb{R}^{n_x} : g(x, \theta) \leq 0\} \quad (5)$$

where $g : \mathbb{R}^{n_x} \times \mathbb{R}^{n_\theta} \rightarrow \mathbb{R}$. In (5), θ represents a parameterization of the class of admissible reachable set estimators: to fix a value of θ is to choose an estimator.

We next fix a functional $\text{Vol} : \mathbb{R}^{n_\theta} \rightarrow \mathbb{R}$ that represents the volume of $\mathcal{R}(\theta)$. This motivates the following scenario program:

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && \text{Vol}(\theta) \\ & \text{subject to} && \theta \in \bigcap_{i=1, \dots, N} \{\theta_i : g(\delta^{(i)}, \theta_i) \leq 0\} \end{aligned} \quad (6)$$

The violation probability, $V(\mathcal{R}(\theta))$, of (6) may be interpreted as the probability that an unseen scenario will violate the bounds of the reachable set estimate. Our goal is to select θ such that the probability of $V(\mathcal{R}(\theta)) > \epsilon$ is less than or equal to β while minimizing $\text{Vol}(\theta)$.

The proposed problem (6) can be equivalently expressed in the functional form

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && \text{Vol}(\theta) \\ & \text{subject to} && g(\delta^{(i)}, \theta) \leq 0, i = 1, \dots, N \\ & && \theta \in \mathbb{R}^{n_\theta}. \end{aligned} \quad (7)$$

The solution to (7) is the minimum-volume set that contains sample points $\delta^{(1)}, \dots, \delta^{(N)}$, and guarantees $\mathbf{P}\{V(\mathcal{R}(\theta)) > \epsilon\} \leq \beta$. The algorithms we present in this section solve (7) given arbitrary values of $\epsilon, \beta \in (0, 1)$ and N samples.

3.1. Tiling with Basis Functions

We first present a method to construct a sublevel set function $g(x, \theta)$ that is convex in θ but nonconvex in x , as was done in (Devonport, 2023). While convex scenario optimization methods can be used to analyze this approach, they require large sample sizes and are not computationally efficient. We show that by utilizing the nonconvex scenario optimization tools introduced in Section 2, we can significantly improve upon these limitations. To construct $g(x, \theta)$, select a finite set of basis functions $f_1(x), \dots, f_m(x) : \mathbb{R}^D \rightarrow \mathbb{R}$ and take g to be

$$g(x, \theta) = \sum_{i=1}^m \theta_i f_i(x) \quad (8)$$

In this section, we will use this approach to construct a *tiling* of the state space and estimate the reachable set of a given problem. To create this *tiling*, assume that the reachable set, $\mathcal{R}$, is contained in a subset $A \subseteq \mathbb{R}^D$. We then partition A into m cells, creating a collection of sets $A_1, \dots, A_m$ such that $\bigcup_{i=1}^m A_i = A$ and $A_i \cap A_j = \emptyset \forall i, j$. This approach can produce arbitrarily fine estimates of the reachable set, depending on how refined the partition is. The accuracy of the partition increases as m increases. Further, we define $\mathbb{1}_{A_i}$ to be the zero-one indicator function for the set A_i , so that $\mathbb{1}_{A_i}(x) = 1$ if $x \in A_i$ and $\mathbb{1}_{A_i}(x) = 0$ otherwise. Therefore, the reachable set estimate is a union of the partitioned cells. We write this as a constrained optimization problem:

$$\begin{aligned} & \underset{\theta}{\text{minimize}} && - \sum_{i=1}^m \theta_i \\ & \text{subject to} && \sum_{i=1}^m \theta_i \mathbb{1}_{A_i}(\delta^{(j)}) \leq 0, \quad j = 1, \dots, N \\ & && \theta \in [0, 1]^m, \end{aligned} \quad (9)$$

To satisfy the scenario constraints, we set $\theta_i = 0 \forall i$ such that scenario $\delta^{(j)} \in A_i$ for at least one $j \in \{1, \dots, N\}$. To minimize the objective while respecting $\theta \in [0, 1]^m$ we set $\theta_i = 1$ for all other A_i . Therefore, $\mathcal{R}(\theta)$ is the solution to (9), the union of cells A_i that contain one or more scenarios $\delta^{(j)}$. This is the minimum volume union of cells that contains all scenarios, $\delta^{(1)}, \dots, \delta^{(N)}$.

We define a support scenario to be the first scenario, $\delta^{(j)}$, in any cell A_i . While this set of support scenarios is not unique, it is irreducible. These scenarios represent the smallest set of reachable states that is possible without changing the solution to (9). This satisfies the conditions needed to obtain the number of support scenarios, s_N^* . We calculate ϵ a-posteriori using (4) with s_N^* and the number of optimization variables, d (the number of cells in our partition). If the obtained ϵ does not satisfy the necessary bounds, we iteratively increase the number of samples, N , and repeat the process. This reachability algorithm based on partition $A_1, \dots, A_m$ is described in Algorithm 1.

Algorithm 1 : Scenario reachability through tiling

- 1: **Input**: Black-box transition function model $\Phi(t_1; t_0, x_0, d)$; Random variables X_0 and D ; Partition dimension m ; Batch size B ; Confidence parameter $\beta \in (0, 1)$.
 - 2: **Output**: $\theta_1, \dots, \theta_m$ corresponding to union of cells A_j such that $A_j \in \mathcal{R}(\theta)$ iff $\theta = 0$; Robustness against constraint violation ϵ .
 - 3: **Initialize** $\theta_j = 1$
 - 4: **while** ϵ is large **do**
 - 5: $N = (B \cdot \text{number of iterations})$
 - 6: **for all** $i \in \{1, \dots, N\}$ **do**
 - 7: Take samples $x_{0i} \sim X_0, d_i \sim D$
 - 8: Evaluate $\delta^{(i)} = \phi(t_1; t_0, x_{0i}, d_i)$
 - 9: If $\delta^{(i)} \in A_i$, then set $\theta_i = 0$
 - 10: **calculate** ϵ using Equation 4 where $s_N^* = |\mathcal{R}(\theta)|$ and $d = m$.
 - 11: **return** $\theta_1, \dots, \theta_m; \epsilon$
-

3.2. Radial Basis Functions (RBFs)

We now turn to a method that allows nonconvexity in both parameters, θ and x . Unlike the method proposed in Section 3.1, this method is not amenable to existing approaches, such as those presented in (Devonport and Arcak, 2020b; Devonport, 2023). In this approach, we construct $g(x, \theta)$ from a finite set of RBFs. We define a RBF, $f(x, \mu, \sigma)$, to be a Gaussian function of x, μ, σ , such that

$$f(x, \mu, \sigma) = e^{-\frac{1}{2} \frac{(x - \mu_i)^2}{\sigma_i^2}} \quad (10)$$

where μ is the center of a RBF and σ is the width of a RBF. We take g to be

$$g(x, \theta) = \sum_{i=1}^m f(x, \mu_i, \sigma_i) - \gamma \quad (11)$$

Therefore, $\theta = (\mu_1, \dots, \mu_m; \sigma_1, \dots, \sigma_m; \gamma)$.

Figure 1 demonstrates that RBFs are particularly well-suited for constructing reachable sets due to the tail interactions that allow multiple RBFs to connect into shapes more complicated than unions of ellipsoids. This approach allows the number of RBFs, m , to be arbitrarily set. If m is larger than

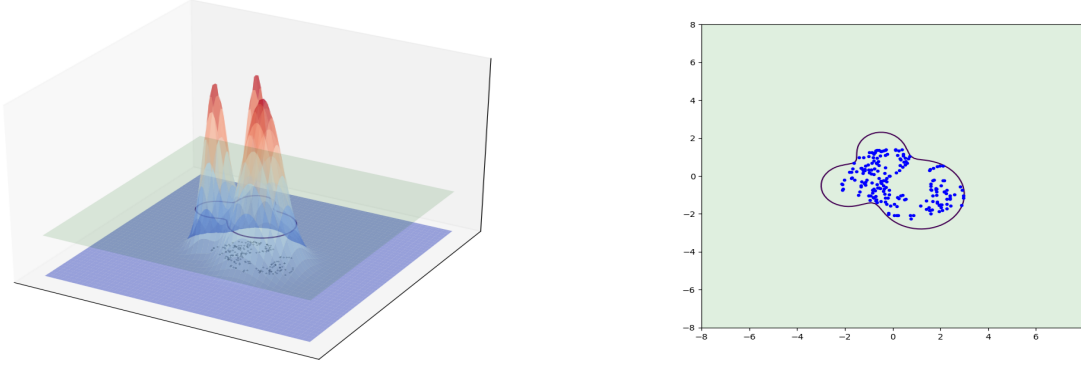


Figure 1: (Left) Radial basis function generated by a set of data with the plane defined by the threshold value, γ , (Right) the contour line (black), created by the intersection of the threshold plane with the RBF, and the set of data (dark blue).

needed, Algorithm 2 will produce disconnected sets. However, the unnecessary RBFs will only contain their initial center. This approach further allows us to produce disconnected reachable sets for systems that exhibit these dynamics. We write this as a constrained optimization problem:

$$\begin{aligned}
 & \underset{\mu, \sigma}{\text{minimize}} && \sum_{i=1}^m \sigma_i^2 \\
 & \text{subject to} && \sum_{i=1}^m e^{-\frac{1}{2} \frac{(\delta^{(j)} - \mu_i)^2}{\sigma_i^2}} - \gamma \geq 0, \quad j = 1, \dots, N, \\
 & && \sigma \in [0, \infty)^m
 \end{aligned} \tag{12}$$

We define a support scenario as any scenario, $\delta^{(j)}$, that changes the solution to Equation (12) when removed from the set of all scenarios, $\delta^{(1)}, \dots, \delta^{(N)}$. In other words, the removal of $\delta^{(j)}$ results in different optimal widths, σ , of the calculated RBFs. It is worth noting that the process of finding these support scenarios can be parallelized. By definition, this satisfies the conditions needed to obtain the number of support scenarios, s_N^* . Due to the nonconvex nature of this problem, we calculate ϵ a-posteriori using (2) with s_N^* . If the obtained value of ϵ does not satisfy the necessary bounds, we iteratively increase the number of samples, N , and repeat the process. This description is summarized in Algorithm 2.

4. Examples

This section demonstrates the ability of the presented data-driven approaches to accurately estimate the forward reachable sets of two numerical examples posed in (Devonport et al., 2021). We evaluate Algorithms 1 and 2, to expose extensions of scenario optimization, but we do not attempt to compare these methods. All computations were done on a Apple M2 Pro, 12-core CPU. One Python thread was run to compute all tiling problems, while ten Python threads were used in computing the radial basis functions.

To verify that the computed reachable sets in the following examples satisfy the guarantee that they are ϵ -accurate, we compute an a-posteriori empirical estimate of the reachable set using a

Algorithm 2 : Scenario reachability with radial basis functions

- 1: **Input:** Black-box transition function model $\phi(t_1; t_0, x_0, d)$; Random variables X_0 and D ; Threshold Γ ; Number of RBFs m ; Batch size B ; Confidence parameter $\beta \in (0, 1)$.
 - 2: **Output:** $\Sigma = \{\sigma_1, \dots, \sigma_m\}$ corresponding to optimal widths of RBFs; $\mu_i, \dots, \mu_m$ corresponding to centers of RBFs; Robustness against constraint violation ϵ .
 - 3: **Initialize** number of support scenarios, $S = 0$.
 - 4: **while** ϵ is large **do**
 - 5: $N = (B \cdot \text{number of iterations})$
 - 6: **Step 1:**
 - 7: **for all** $i \in \{1, \dots, N\}$ **do**
 - 8: Take samples $x_{0i} \sim X_0, d_i \sim D$
 - 9: Evaluate $\delta^{(i)} = \phi(t_1; t_0, x_{0i}, d_i)$
 - 10: Take $\gamma = \Gamma$; $\mu_1, \dots, \mu_m = \text{k-means}(m \text{ clusters})$; and $\sigma_1, \dots, \sigma_m$ to be arbitrary
 - 11: Evaluate $c = e^{-\frac{1}{2} \frac{(\delta^{(i)} - \mu_i)^2}{\sigma_i^2}} - \gamma$
 - 12: If $c \geq 0$, then set $\sigma_i = 0$. Else set $\sigma_i = \infty$.
 - 13: Let $\Sigma = \{\sigma_1, \dots, \sigma_m\}$ be the set of optimal widths of the calculated RBFs from Step 1.
 - 14: **Step 2:**
 - 15: **for all** $i \in \{1, \dots, N\}$ **do**
 - 16: Remove sample $\delta^{(i)}$ from $\delta^{(1)}, \dots, \delta^{(N)}$ taken from Step 1
 - 17: **for all** $j \in \{1, \dots, N\} \setminus i$ **do**
 - 18: Evaluate $c = e^{-\frac{1}{2} \frac{(\delta^{(j)} - \mu_j)^2}{\sigma_j^2}} - \gamma$
 - 19: If $c \geq 0$, then set $\sigma_j = 0$. Else set $\sigma_j = \infty$.
 - 20: Let $\Sigma_i = \{\sigma_1, \dots, \sigma_m\}$ be the set of optimal widths of the RBFs without scenario $\delta^{(i)}$.
 - 21: If $\Sigma_i = \Sigma$, then $S = S + 1$.
 - 22: **calculate** ϵ using Equation 2 where $s_N^* = S$.
 - 23: **return** $\Sigma; \mu_i, \dots, \mu_m; \epsilon$.
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one-sided Chernoff bound (Tempo et al., 2012). This ensures that the a-posteriori estimate of the reachable set given an additional 46,052 samples exceeds the true measure by no more than .01 with confidence 0.9999. The results are shown in Tables 1 and 2 and validate that the reachable sets are indeed ϵ -accurate.

4.1. Duffing Oscillator

The first example is a reachable set estimation problem for the nonlinear, time-varying system with dynamics: $\ddot{x} = -\alpha y + x - x^3 + \gamma \cos(\omega t)$, with states $x, y \in \mathbb{R}$ and parameters $\alpha, \gamma, \omega \in \mathbb{R}$. This system is known as the Duffing oscillator, a nonlinear oscillator which exhibits chaotic behavior for certain values of α, γ and ω , for instance $\alpha = 0.05, \gamma = 0.4, \omega = 1.3$. The set of initial states is the interval such that $x(0) \in [0.95, 1.05]$, $y(0) \in [-0.05, 0.05]$, and we take X_0 to be the uniform random variable over this interval. The time range is $[t_0, t_1] = [0, 100]$.

4.1.1. TILING WITH BASIS FUNCTIONS

We take A to be the hyperrectangle $A = [-5, 5] \times [-5, 5]$ and partition A into a 20x20 grid. We take $N = 1000$ samples and $\beta = 10^{-9}$. The output of Algorithm 1 for this problem, which took 1.70

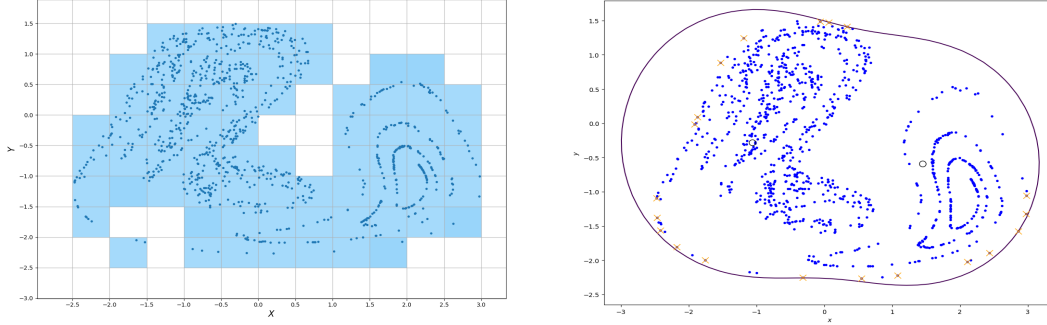


Figure 2: Duffing Oscillator problem in $\mathbb{R}^2$: (Left) Enlarged view of reachable set (light blue) using a grid-based partition and $N=1000$ samples (dark blue). (Right) Reachable set (purple) and support scenarios (orange) using two radial basis functions and $N=1000$ samples (dark blue).

seconds to compute, is shown on the left in Figure 2. We calculate the number of support scenarios, $s_N^* = 67$, and $\epsilon = 0.2509$ a-posteriori to get the probability that our tiling contains at least 74.91% of the reachable set distribution, with a "one in a billion" chance of failure. We performed 100 trials of the same experiment and the number of support scenarios was always between 63 and 72. On average, the probabilistic volume outside the reachable set did not exceed 25.09% under the convexity refined nonconvex approach. Additionally, we utilize the wait-and-judge perspective to investigate the effect of a larger N on ϵ . The results are shown in Table 1.

4.1.2. RADIAL BASIS FUNCTIONS

We take $m = 2$, such that we will calculate two radial basis functions. We set the initial centers of the RBFs to be close to optimal using a k-means clustering algorithm and allow the widths of our RBF to be chosen arbitrarily. We take $\gamma = 0.25$ to be the threshold of our RBF, $N = 1000$ samples, and $\beta = 10^{-9}$. The output of Algorithm 2 for this problem, which took approximately 7 minutes to compute, is shown on the right in Figure 2. We calculate the number of support scenarios, $s_N^* = 19$, and $\epsilon = 0.1182$ a-posteriori to get the probability that our radial basis functions contain at least 88.82% of the reachable set distribution, with a "one in a billion" chance of failure. We performed 100 trials of the same experiment and the number of support scenarios was always between 5 and 24. On average, the probabilistic volume outside the reachable set did not exceed 10.34% under the nonconvex approach. Additionally, we utilize the wait-and-judge perspective to investigate the effect of a larger N on ϵ . The results are shown in Table 1.

4.2. Quadrotor Model

The next example is a reachable set estimation problem for a nonlinear model of a quadrotor used as an example in (Mitchell et al., 2019) and (Bouffard, 2012). The dynamics for this system are

$$\ddot{x} = u_1 K \sin(\theta), \quad \ddot{h} = -g + u_1 K \cos(\theta), \quad \ddot{\theta} = -d_0 \theta - d_1 \dot{\theta} + n_0 u_2 \quad (13)$$

where x and h denote the quadrotor's horizontal position and altitude in meters, respectively, and θ denotes its angular displacement. The system has 6 states, which we take to be x, h, θ , and their first derivatives. The two system inputs u_1 and u_2 represent the motor thrust and the desired

	Sample Size	Run Time of (1)/(2)	ϵ	Estimate
Duffing Oscillator: Tiling	N = 1000	1.70s	0.251	.999966
	N = 3000	2.49s	0.117	
	N = 5000	3.48s	0.078	
Duffing Oscillator: RBF	N = 1000	7 min	0.118	.999971
	N = 2000	11 min	0.033	
	N = 3000	22 min	0.035	

Table 1: Computation times and a-posteriori calculation of ϵ for reachable sets of N samples of Duffing Oscillator, and empirical estimates of the reachable sets (in Figure 2) using a one-sided Chernoff bound.

angle, respectively. The parameter values used (following (Bouffard, 2012)) are $g = 9.81$, $K = 0.89/1.4$, $d_0 = 70$, $d_1 = 17$, $n_0 = 55$. The set of initial states is the interval such that

$$\begin{aligned} x(0) &\in [-1.7, 1.7], & h(0) &\in [0.3, 2.0], & \theta(0) &\in [-\pi/12, \pi/12], \\ \dot{x}(0) &\in [-0.8, 0.8], & \dot{h}(0) &\in [-1.0, 1.0], & \dot{\theta}(0) &\in [-\pi/2, \pi/2], \end{aligned} \quad (14)$$

the set of inputs is the set of constant functions $u_1(t) = u_1$, $u_2(t) = u_2 \forall t \in [t_0, t_1]$, whose values lie in the interval $u_1 \in [-1.5 + g/K, 1.5 + g/K]$, $u_2 \in [-\pi/4, \pi/4]$, and we take X_0 and D to be the uniform random variables defined over these intervals. The time range is $[t_0, t_1] = [0, 5]$.

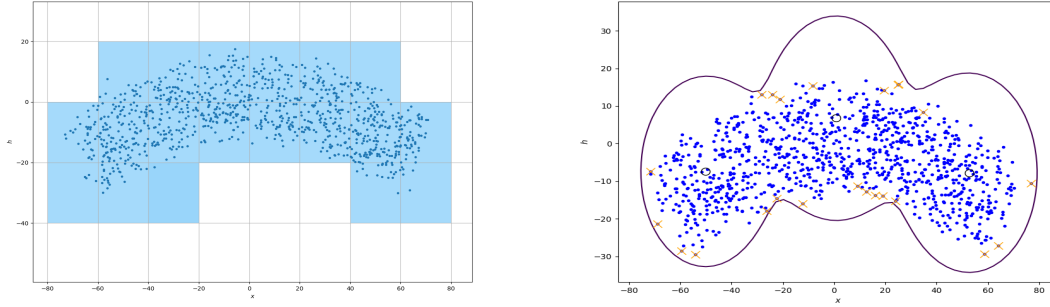


Figure 3: Reachable set estimates for the horizontal position and altitude of the planar quadrotor model: (Left) Enlarged view of reachable set (light blue) using a grid-based partition and $N=1000$ samples (dark blue). (Right) Reachable set (purple) and support scenarios (orange) using three radial basis functions and $N=1000$ samples (dark blue).

4.2.1. TILING WITH BASIS FUNCTIONS

We take A to be the hyperrectangle $A = [-100, 100]^6$ and partition A to be the grid with 10 sides along each dimension. We take $N = 1000$ samples and $\beta = 10^{-9}$. The output of Algorithm 1 for this problem, which took 6.28 seconds to compute, is shown on the left in Figure 3. We calculate the number of support scenarios, $s_N^* = 19$, and $\epsilon = 0.1125$ a-posteriori to get the probability that our tiling contains at least 88.75% of the reachable set distribution, with a "one in a billion" chance

of failure. We performed 100 trials of the same experiment and the number of support scenarios remained between 18 and 22. On average, the probabilistic volume outside the reachable set did not exceed 11.6% under the convexity refined nonconvex approach. Additionally, we investigate the effect of a larger N on ϵ . The results are shown in Table 2.

	Sample Size	Run Time of (1)/(2)	ϵ	Estimate
Quadrotor: Tiling	N = 1000	6.28 s	0.113	.999997
	N = 3000	8.07 s	0.050	
	N = 5000	8.37 s	0.032	
Quadrotor: RBF	N = 1000	30 min	0.125	.999923
	N = 2000	3.3 hr	0.065	
	N = 3000	5.5 hr	0.051	

Table 2: Computation times and a-posteriori calculation of ϵ for reachable sets of N samples of Quadrotor, and empirical estimates of the reachable sets (in Figure 3) using a one-sided Chernoff bound.

4.2.2. RADIAL BASIS FUNCTIONS

We take $m = 3$, such that we will calculate three radial basis functions. We set the initial centers of the RBFs to be close to optimal using a k-means clustering algorithm and allow the widths of our RBF to be chosen arbitrarily. We take $\gamma = 0.25$ to be the threshold of our RBF, $N = 1000$ samples, and $\beta = 10^{-9}$. The output of Algorithm 2 for this problem, which took approximately 30 minutes to compute, is shown on the right in Figure 3. We calculate the number of support scenarios, $s_N^* = 22$, and $\epsilon = 0.1253$ a-posteriori to get the probability that our radial basis functions contain at least 87.47% of the reachable set distribution, with a "one in a billion" chance of failure. We performed 100 trials of the same experiment and the number of support scenarios was always between 14 and 31. On average, the probabilistic volume outside the reachable set did not exceed 11.8% under the nonconvex approach. Additionally, we utilize the wait-and-judge perspective to investigate the effect of a larger N on ϵ . The results are shown in Table 2.

5. Conclusion

We presented a method of nonconvex scenario optimization for reachability analysis. This approach does not require a-priori sample complexity bounds and significantly decreases the computational cost in finding reachable sets. We first provided a partition-based instance of scenario reachability that is computationally efficient and scales well to higher state dimensions. We then provided an estimation of reachable sets using radial basis functions.

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